

UNIVERSITÀ DEGLI STUDI DI TORINO

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Category theory

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Table of Contents

1	Theory	3
1.1	The pullback functor	3

Introduction

Textt

Chapter 1

Theory

1.1 The pullback functor

In this section we will define the pullback functor and its left adjoint, the shrink functor. Given a category $\mathcal{A}$ with pullbacks and a morphism $p : E \rightarrow A$ in $\mathcal{A}$, one can consider adjunction

$$(\mathcal{A} \downarrow B) \quad \begin{array}{c} \xleftarrow{p^*} \\ \top \\ \xrightarrow{p!} \end{array} \quad (\mathcal{A} \downarrow E)$$

defined as follows:

- $(\mathcal{A} \downarrow B)$ is the category of objects over B in $\mathcal{A}$, we denote an object by $(A, \alpha : A \rightarrow B)$;
- $(\mathcal{A} \downarrow E)$ is the category of objects over E in $\mathcal{A}$, we denote an object by $(D, \delta : D \rightarrow E)$;
- $p^* : (\mathcal{A} \downarrow E) \rightarrow (\mathcal{A} \downarrow B)$ is the pullback functor

$$\begin{array}{ccc}
 \begin{array}{c} A \\ \downarrow \alpha \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{ccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & & \\ \downarrow \text{proj}_1 & & \downarrow \alpha & & \\ E & \xrightarrow{p} & B & & \end{array} \\
 & & \exists! \tilde{f} & & \\
 \begin{array}{ccc} A & \xrightarrow{f} & A' \\ \downarrow \alpha & \searrow \alpha' & \\ B & & \end{array} & \xrightarrow{p^*} & \begin{array}{ccccccc} E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}'_2} & E \times_B A' \\ & \searrow \alpha & \searrow \alpha' & & \swarrow & & \swarrow \\ & & B & & & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}'_1 & & \\ & E & & & & & \end{array}
 \end{array}$$

- $p_! : (\mathcal{A} \downarrow B) \rightarrow (\mathcal{A} \downarrow E)$ is the shrink functor

$$\begin{array}{ccc}
 \begin{array}{c} D \\ \downarrow \delta \\ E \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & & \\ \delta \downarrow & \searrow p\delta & \\ E & \xrightarrow{p} & B \end{array} \\
 \\
 \begin{array}{ccc} D & \xrightarrow{d} & D' \\ \delta \downarrow & \swarrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{d} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ E & \xrightarrow{p} & B \end{array}
 \end{array}$$

- the unit η^p

$$\begin{array}{l}
 \eta : \text{Id}_{(\mathcal{A} \downarrow E)} \rightarrow p^* p_! \\
 \eta_{(D, \delta)} = \langle \delta, 1_D \rangle : (D, \delta) \rightarrow (E \times_B D, \text{proj}_1)
 \end{array}$$

$$\begin{array}{ccccc}
 D & & \xrightarrow{1_D} & & D \\
 \delta \downarrow & \searrow \exists! \langle \delta, 1_D \rangle & & & \downarrow p\delta \\
 E \times_B D & \xrightarrow{\quad} & D & & \\
 \downarrow & & \downarrow p & & \\
 E & \xrightarrow{p} & B & &
 \end{array}$$

- the counit ϵ^p

$$\begin{array}{l}
 \epsilon : p_! p^* \rightarrow \text{Id}_{(\mathcal{A} \downarrow B)} \\
 \epsilon_{(A, \alpha)} = \text{proj}_2 : (E \times_B A, p(\text{proj}_1)) \rightarrow (A, \alpha)
 \end{array}$$

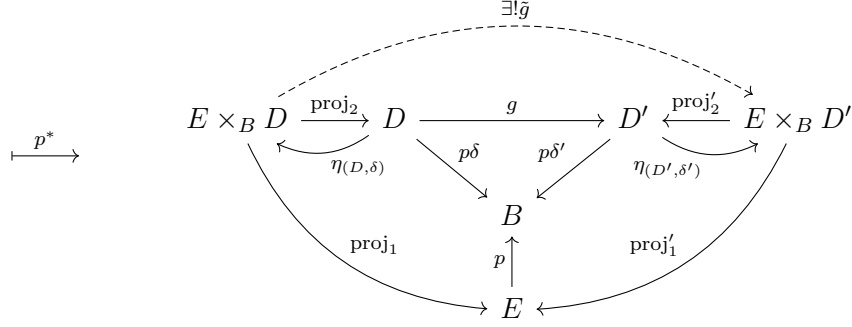
$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\text{proj}_2} & A \\
 \downarrow \text{proj}_1 & & \downarrow \alpha \\
 E & \xrightarrow{p} & B
 \end{array}$$

Let us show that this is actually an adjunction. First we want to prove that the unit is a natural transformation. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D \\
 \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} \\
 D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D'
 \end{array}$$

In order to do that we need to expand the action of the functor $p^* p_!$ on the morphism g . We have:

$$\begin{array}{ccc}
 \begin{array}{ccc} D & \xrightarrow{g} & D' \\ \delta \downarrow & \swarrow \delta' & \\ E & & \end{array} & \xrightarrow{p_!} & \begin{array}{ccc} D & \xrightarrow{g} & D' \\ & \searrow p\delta & \downarrow p\delta' \\ & & B \end{array}
 \end{array}$$



Considering the diagram

$$\begin{array}{ccccc}
 D & \xrightarrow{\eta_{(D,\delta)}} & E \times_B D & & \\
 \downarrow g & \searrow \text{dashed} & \downarrow p^*p_!(g)=\tilde{g} & & \\
 D' & \xrightarrow{\eta_{(D',\delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\
 & & \downarrow \text{proj}'_1 & & \downarrow p\delta' \\
 & & E & \xrightarrow{p} & B
 \end{array}$$

we can prove the naturality in the left diagram by the universal property of the pullback in the right. Since $(\text{proj}_1, \text{proj}_2)$ are jointly monic, by the following compositions

- $(\text{proj}'_2)(\tilde{g})(\eta_{(D,\delta)}) = (g)(\text{proj}_2)(\eta_{(D,\delta)}) = (g)(1_D) = g$
- $(\text{proj}'_2)(\eta_{(D',\delta')})(g) = (1_{D'})(g) = g$
- $(\text{proj}'_1)(\tilde{g})(\eta_{(D,\delta)}) = (\text{proj}_1)(\eta_{(D,\delta)}) = \delta$
- $(\text{proj}'_1)(\eta_{(D',\delta')})(g) = (\delta')(f) = \delta$

we conclude that $(\tilde{g})(\eta_{(D,\delta)}) = (\eta_{(D',\delta')})(g)$. So the naturality condition of η is satisfied.

Now we want to prove that the counit is a natural transformation. Consider $f : (A, \alpha) \rightarrow (A', \alpha')$ a morphism in $(\mathcal{A} \downarrow B)$, we want to prove that the following diagram commutes:

$$\begin{array}{ccc}
 E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)}} & A \\
 p_!p^*(f)=\tilde{f} \downarrow & & \downarrow f \\
 E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')}} & A'
 \end{array}$$

As similarly done above, we need to expand the action of the functor $p_!p^*$ on the morphism f , we have:

$$\begin{array}{ccc}
 \begin{array}{c} A \xrightarrow{f} A' \\ \downarrow \alpha \quad \swarrow \alpha' \\ B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & & \curvearrowleft & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \downarrow & \swarrow \alpha' & \\ & & B & & \\ \text{proj}_1 \swarrow & & \uparrow p & & \searrow \text{proj}_1' \\ & & E & & \end{array} \end{array} \\
 & & \xrightarrow{p!} & \begin{array}{c} \begin{array}{ccccc} & & \exists! \tilde{f} & & \\ & \curvearrowright & & \curvearrowleft & \\ E \times_B A & \xrightarrow{\text{proj}_2} & A & \xrightarrow{f} & A' & \xleftarrow{\text{proj}_2'} & E \times_B A' \\ & \searrow \alpha & \downarrow & \swarrow \alpha' & \\ & & B & & \\ p(\text{proj}_1) \swarrow & & \uparrow & & \searrow p(\text{proj}_1') \\ & & & & \end{array} \end{array}
 \end{array}$$

Since $\epsilon_{(A,\alpha)}$ is the $\text{proj}_{2,E \times_B A}$, the composition $(f)(\epsilon_{(A,\alpha)}) = (\epsilon_{(A',\alpha')})(\tilde{f})$ holds by the definition of $\tilde{f}$. So also the naturality of ϵ is satisfied.

Now we can check that the unit and counit fulfil the first triangle identities. Let (A, α) be an object in $(\mathcal{A} \downarrow B)$, we have:

$$((p^* \epsilon) \circ (\eta p^*))_{(A,\alpha)} = (p^* \epsilon)_{(A,\alpha)} (\eta p^*)_{(A,\alpha)} = p^*(\text{proj}_{2,E \times_B A})(\eta_{(E \times_B A, \text{proj}_1)}) = \circledast$$

Considering first how p^* acts on the morphism $\text{proj}_{1,E \times_B A}$

$$\begin{array}{ccc}
 \begin{array}{ccc} E \times_B A & \xrightarrow{\text{proj}_{1,E \times_B A}} & A \\ & \searrow \alpha(\text{proj}_{2,E \times_B A}) & \downarrow \alpha \\ & & B \end{array} & \xrightarrow{p^*} & \begin{array}{c} \text{proj}_{2,E \times_B A} \\ \curvearrowright \\ E \times_B (E \times_B A) \xrightarrow{\text{proj}_{2,E \times_B (E \times_B A)}} E \times_B A \xrightarrow{\text{proj}_{2,E \times_B A}} A \xleftarrow{\text{proj}_{2,E \times_B A}} E \times_B A \\ \quad \quad \quad \searrow (\alpha)(\text{proj}_{2,E \times_B A}) \quad \downarrow \alpha \quad \swarrow \\ \quad \quad \quad \quad \quad B \\ \quad \quad \quad \text{proj}_{1,E \times_B (E \times_B A)} \quad \uparrow p \quad \quad \text{proj}_{1,E \times_B A} \\ \quad \quad \quad \quad \quad E \end{array}
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\eta_{(E \times_B A, \text{proj}_1)})$$

and then considering the definition of η , we obtain

$$\begin{array}{ccccc}
 & & 1_{E \times_B A} & & \\
 & \nearrow & & \searrow & \\
 E \times_B A & & & & E \times_B A \\
 & \searrow \exists! \langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle & & \nearrow & \\
 & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \\
 & \downarrow & & \downarrow p(\text{proj}_{1, E \times_B A}) & \\
 \text{proj}_{1, E \times_B A} \searrow & E & \xrightarrow{p} & B & \\
 & & & &
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle)$$

In order to conclude we need to check some compositions so that, by the universal property of the pullback $E \times_B A$ in the right of the following diagram, we obtain the identity morphism.

$$\begin{array}{ccccccc}
 & & 1_{E \times_B A} & & \tilde{\text{proj}}_2 & & \\
 & \nearrow & & \searrow & & \nearrow & \\
 E \times_B A & \xrightarrow{\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle} & E \times_B (E \times_B A) & \longrightarrow & E \times_B A & \xrightarrow{\text{proj}_{2, E \times_B A}} & A \longleftarrow E \times_B A \\
 & \searrow & \downarrow & & \downarrow (\alpha)(\text{proj}_{2, E \times_B A}) & \nearrow \alpha & \\
 & & & & B & & \\
 & \searrow \text{proj}_{1, E \times_B (E \times_B A)} & & & \uparrow p & \searrow \text{proj}_{1, E \times_B A} & \\
 & & E & & & &
 \end{array}$$

$$\circledast = (\tilde{\text{proj}}_2)(\langle \text{proj}_{1, E \times_B A}, 1_{E \times_B A} \rangle) = \text{Id}_{(E \times_B A, \text{proj}_1)}$$

We want to check that composition of the projections of $E \times_B A$ with $\circledast$ is the same of composing those projection with $\text{Id}_{(E \times_B A, \text{proj}_1)}$. Following the first composition

$$\begin{aligned}
 (\text{proj}_{2, E \times_B A})(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_{2, E \times_B A})(\text{proj}_{2, E \times_B (E \times_B A)})(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\
 &= (\text{proj}_{2, E \times_B A})(1_{E \times_B A}) = \text{proj}_{2, E \times_B A} = (\text{proj}_{2, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

and the second composition

$$\begin{aligned}
 (\text{proj}_{1, E \times_B A})(\tilde{\text{proj}}_2)(\langle \text{proj}_1, 1_{E \times_B A} \rangle) &= (\text{proj}_{1, E \times_B (E \times_B A)})(\langle \text{proj}_1, 1_{E \times_B A} \rangle) = \\
 \text{proj}_{1, E \times_B A} &= (\text{proj}_{1, E \times_B A})(\text{Id}_{(E \times_B A, \text{proj}_1)})
 \end{aligned}$$

Finally we can check the second triangle identities. Let (D, δ) be an object in $(\mathcal{A} \downarrow E)$, we have:

$$\begin{aligned} ((\epsilon p_!) \circ (p_! \eta))_{(D, \delta)} &= ((\epsilon p_!)_{(D, \delta)} (p_! \eta))_{(D, \delta)} = (\epsilon_{p_!(D, \delta)} p_!(\langle \delta, 1_D \rangle)) = \\ &= (\epsilon_{(D, p\delta)} p_!(\langle \delta, 1_D \rangle)) = \text{proj}_{2, E \times_B D}(\langle \delta, 1_D \rangle) = 1_D = \text{Id}_{(D, \delta)} \end{aligned}$$

$$\begin{array}{ccc} E \times_B D & \xrightarrow{\text{proj}_{2, E \times_B D}} & D \\ \downarrow \text{proj}_{1, E \times_B D} & & \downarrow p\delta \\ E & \xrightarrow{p} & B \end{array}$$

So we can conclude that the adjunction holds.

$$\begin{array}{ccc} & p^* & \\ (\mathcal{A} \downarrow B) & \top & (\mathcal{A} \downarrow E) \\ & p_! & \end{array}$$

Remark 1.1. p^* is a right adjoint functor, so it preserves limits. $p_!$ is a left adjoint functor, so it preserves colimits.

Now we can claim some properties of the adjunction.

Property 1.2. The unit η and the counit ϵ are cartesian.

Proof. In order to prove that η is cartesian, ie the naturality diagrams are pullback. Let $g : (D, \delta) \rightarrow (D', \delta')$ be a morphism in $(\mathcal{A} \downarrow E)$. Considering the commutative diagram below, since both the bottom and the total diagrams are pullback, it follows that the top diagram is a pullback.

$$\begin{array}{ccccc} E \times_B D & \xrightarrow{\text{proj}_2} & D & & \\ \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g & & \\ E \times_B D' & \xrightarrow{\text{proj}'_2} & D' & & \\ \downarrow \text{proj}'_1 & & \downarrow p\delta' & & \\ E & \xrightarrow{p} & B & & \end{array}$$

(Left curved arrow: proj_1 , Right curved arrow: $p\delta$)

Considering the following commutative diagram, since the right diagram is a pullback, as we just proved, and the total diagram is a pullback, we can conclude that the left diagram is a pullback, as stated.

$$\begin{array}{ccccc} & & 1_D & & \\ & \searrow & \downarrow & \searrow & \\ D & \xrightarrow{\eta_{(D, \delta)}} & E \times_B D & \xrightarrow{\text{proj}_2} & D \\ \downarrow g & & \downarrow p^* p_!(g) = \tilde{g} & & \downarrow g \\ D' & \xrightarrow{\eta_{(D', \delta')}} & E \times_B D' & \xrightarrow{\text{proj}'_2} & D' \\ & \searrow & \downarrow & \searrow & \\ & & 1_{D'} & & \end{array}$$

$$\begin{array}{ccc}
E \times_B A & \xrightarrow{\epsilon_{(A,\alpha)} = \text{proj}_2} & A \\
\downarrow p_! p^*(f) = \bar{f} & & \downarrow f \\
E \times_B A' & \xrightarrow{\epsilon_{(A',\alpha')} = \text{proj}_2'} & A' \\
\downarrow \text{proj}_1' & & \downarrow \alpha' \\
E & \xrightarrow{p} & B
\end{array}
\quad \begin{array}{l} \text{proj}_1 \curvearrowright \\ \curvearrowleft \alpha \end{array}$$

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